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Lab 4: Implementation and Comparison of Agile Project Management Tools – JIRA, Asana, Trello, ClickUp, and Zoho

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OBJECTIVE:

Implementation and Comparison of Agile Project Management Tools – JIRA, Asana, Trello, ClickUp, and Zoho

AIM :

To explore, compare, and implement Agile project management tools — JIRA, Asana, Trello, ClickUp, and Zoho Projects — for planning sprints, managing tasks, and tracking progress in Agile workflows.

THEORY :

Agile project management tools provide digital Kanban boards, sprint planners, issue trackers, and collaboration features that support Scrum and Kanban methodologies.

- JIRA (Atlassian): Industry-standard tool for software teams. Supports Scrum boards, Kanban, epics, stories, sprints, and backlog management.
- Asana: Task and project management with timeline (Gantt), board, and list views. Suited for both technical and non-technical teams.
- Trello: Simple Kanban board tool using cards and columns. Ideal for small teams and straightforward workflows.
- ClickUp: All-in-one productivity tool combining tasks, docs, goals, and time tracking with multiple views (list, board, Gantt, calendar).
- Zoho Projects: Offers task management, Gantt charts, time logging, and integration with other Zoho apps.

OBJECTIVES :

- Create a project and board in JIRA, Trello, and ClickUp.
- Define a sprint with user stories and tasks.
- Move tasks across workflow stages (To Do, In Progress, Done).
- Compare features across the tools.

ALGORITHM :

1. Create an account on JIRA, Trello, Asana, ClickUp, and Zoho.
2. Create a new project named 'Agile Lab Project' in each tool.
3. Add a board with columns: To Do, In Progress, Review, Done.
4. Create user stories and break them into tasks with due dates.
5. Move tasks through the board stages.
6. Compare sprint velocity, reporting, and ease of use across tools.

CODE :

JIRA – Creating an issue via REST API:

```
curl -X POST https://your-domain.atlassian.net/rest/api/3/issue \
-H "Content-Type: application/json" \
-u email@example.com:API_TOKEN \
-d '{ "fields": { "project": { "key": "LAB" },
"summary": "Implement login feature",
"issuetype": { "name": "Story" } } }'
```

Trello – Adding a card via API:

```
curl -X POST "https://api.trello.com/1/cards" \  
-d "name=Implement+Login&idList=LIST_ID&key=API_KEY&token=TOKEN"
```

OUTPUT :

- JIRA: Sprint board created with 5 user stories across workflow columns.
- Trello: Kanban board set up with cards moved across lists successfully.
- Asana: Timeline view displayed task dependencies clearly.
- ClickUp: Dashboard showed task completion percentage and time estimates.
- Zoho: Gantt chart generated automatically from task due dates.

RESULT :

All five Agile project management tools were successfully configured and used to manage a sample sprint.

CONCLUSION :

Different Agile PM tools serve different team needs. JIRA suits large software teams, Trello suits small teams needing simplicity, ClickUp and Asana balance features with usability, while Zoho integrates well within business suites.